Debt Capacity and the Anatomy of Sovereign Crises

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Sovereign defaults are rarely complete

- Governments in distress keep paying some obligations and suspend others.
- Missed payments accumulate as arrears that add to the debt.
- Countries often leave a default episode owing more than when it began.

Arellano, Mateos-Planas, and Ríos-Rull (2023): a theory of partial default.

- The sovereign chooses what share of its debt service to withhold.
- It never loses market access. The price reflects the expected default intensity.
- Data: in partial default a third of the time, withholding $\sim$40% of debt service.

We regard partial default as the right description of sovereign distress and the natural baseline.

What does partial default teach us about sovereign crises?

Four lessons.

1. Debt capacity. Set by the cost of defaulting, not the value of market access.
2. Entry. Defaults begin with a jump, when shocks overwhelm a precautionary buffer.
3. The episode. A financing regime: borrowing continues until output recovers.
4. Pessimism. Lenders' expectations alone can produce a slow crisis, without a run.

Three versions of the model: deterministic, a Markov chain, and a calibrated diffusion. Tractable, not stylized: the calibrated model fits the data at least as well as theirs.

Why continuous time

- The default intensity is a flow decision. Arrears accrue continuously.
- The sovereign never loses access: no stopping problem, no variational inequality.
- One HJB equation and one linear pricing equation, at every state.
- Given the price, the default decision is static: closed-form thresholds.
- Deterministic model: a pair of ODEs, with closed forms at the steady state.
- Computation: no smoothing shocks, exact moments, minutes per evaluation.
- Fast enough to calibrate all nine free parameters by direct search.

Related literature

- **Partial default and arrears:** Arellano, Mateos-Planas, and Ríos-Rull (2023).
- **Binary default:** Eaton–Gersovitz (1981), Aguiar–Gopinath (2006), Arellano (2008).
- **Long-term debt:** Hatchondo–Martinez (2009), Chatterjee–Eyigungor (2012).
- **Dilution:** Hatchondo–Martinez–Sosa-Padilla (2016), Aguiar–Amador (2020).
- **Continuous time:** Bornstein (2020), Tourre (2017), Aguiar et al. (2019), Hurtado–Nuño–Thomas (2023).
- **No commitment:** DeMarzo–He–Tourre (2023). **Reputation:** Amador–Phelan (2021, 2023), Tsoy (2026).
- **Self-fulfilling crises:** Calvo (1988), Cole–Kehoe (2000), Lorenzoni–Werning (2019), Aguiar et al. (2022).
- **Price floors and spread brakes:** Chatterjee–Eyigungor (2015), Hatchondo–Martinez–Roch (2022).
- **Computation:** Achdou et al. (2022), Rendahl (2022), Mihalache (2025).

The Model

The sovereign

- Preferences $\mathbb{E}_0 \int_0^\infty e^{-\rho t} u(c_t)\, dt$. Log endowment z_t, a Markov process.
- Long-term debt B_t: coupon λ, maturity rate δ, debt service $(\delta + \lambda)B_t$.
- Default intensity $d_t \in [0, 1]$: share withheld. A fraction κ becomes arrears.
- Issuance ℓ_t at the market price q_t. Buybacks allowed.

$$c_t = \phi(d_t, z_t)\, e^{z_t} - (1 - d_t)(\delta + \lambda)B_t + q_t\, \ell_t, \qquad \dot{B}_t = \ell_t - \delta B_t + \kappa(\delta + \lambda)\, d_t B_t$$

- Write $a(d) := \delta - \kappa(\delta + \lambda)d$: absent issuance, debt decays at rate $a(d)$.
- $1 - \kappa$: the haircut lenders suffer on defaulted payments.

The cost of default

$$\phi(d,z) = (1-\gamma_0 d^{\gamma_1})\,F(z) \;\text{ for } d>0, \qquad \phi(0,z)=1,$$
$$F(z) = 1-\gamma_2\,(e^z - y^*)\,\mathbb{1}\{e^z \geq y^*\}$$

- **Variable cost** $\gamma_0 d^{\gamma_1}$: convex, zero marginal cost at $d=0$.
- **Fixed cost** $(1-F(z))e^z$: paid whenever the sovereign defaults at all.
- Larger in good times. Zero below y^*: the *no-fixed-cost region*.
- The jump of ϕ at $d=0$ creates a region of no default.

Same structure as Arellano, Mateos-Planas, and Ríos-Rull.

Lenders and the bond price

- Competitive, risk-neutral lenders with outside option r. Coupon $\lambda = r$, so $q^{\text{rf}} = 1$.
- A unit pays $(1-d)(\delta+\lambda)$ in cash and $\kappa(\delta+\lambda)d$ in new units.

$$q(B,z) = \mathbb{E}_{B,z} \int_0^\infty e^{-\int_0^t \xi(d_s)\, ds} (1-d_t)(\lambda+\delta)\, dt, \qquad \xi(d) := r + \delta - \kappa(\delta+\lambda)\, d$$

- Feynman–Kac: $\xi(d)\, q = (1-d)(\lambda+\delta) + \tilde{S}\, q_B + \mathcal{L}_z q$, linear in q.
- Spread $= (\delta+\lambda)/q - (\delta+r)$.

Equilibrium

Given the price $q(B,z)$, the value solves

$$\rho V(B,z) = \max_{c,\, d \in [0,1]} \{u(c) + S(B,z,c,d,q)\, V_B + \mathcal{L}_z V\}$$

- Markov equilibrium: V, policies (c,d), and price q that solve both equations.
- A single value function. No exclusion regime, no stopping problem.
- Stationary distribution: the adjoint of the same operator, no simulation.
- First-order condition at an interior drift: $u'(c)\, q = -V_B$.

Arrears are a source of funds at a fixed price

Let n_t be the total addition to debt, from issuance and arrears. Then

$$c_t = \phi(d_t, z_t)e^{z_t} - (\delta + \lambda)B_t + q_t\, n_t + (1 - \kappa q_t)(\delta + \lambda)B_t\, d_t.$$

- Withholding a unit of debt service raises consumption by $1 - \kappa q_t$, at an output cost.
- Its price is fixed by the recovery κ, with the haircut falling on existing creditors.
- Its terms do not deteriorate with the sovereign's creditworthiness.
- So arrears are attractive precisely when market prices are low.

Lenders price expected withholding. No transfer: the output cost is a deadweight loss.

The underwriting standard

- Problem: with no fixed cost, full default makes debt nearly free at the margin.
- Lenders still pay a positive price, so the sovereign borrows up to any limit.
- Quantitative models leave the limit implicit, as the top of the debt grid.

Our fix: underwriters do not place new issues below a price floor $\underline{q}$,

$$\ell_t \leq 0 \quad \text{whenever} \quad q(B_t, z_t) < \underline{q}.$$

- Outstanding debt still trades. Buybacks and arrears unchanged.
- Makes the level of the borrowing limit irrelevant. Binds on $<1\%$ of the mass.
- A price floor (Chatterjee–Eyigungor 2015), a spread brake (Hatchondo et al. 2022).

The default decision is static

For a given drift S, consumption is $c = \mathcal{R}(d; B, z, q) + q\,S$, with the *resource function*

$$\mathcal{R}(d; B, z, q) := \phi(d, z)e^{z} - (1 - d)(\delta + \lambda)B + q\,a(d)\,B.$$

Proposition

Wherever the standard is slack, every HJB maximizer sets $d^{*} = \arg\max_{d} \mathcal{R}(d; B, z, q)$.

- Independent of the value function and of where debt is going.
- Marginal gain from withholding: $(1 - \kappa q)(\delta + \lambda)B$, payment saved net of arrears.

Explicit thresholds

Interior intensity and the full-default threshold, at any state and price:

$$d_{\text{int}} = \left[\frac{(1-\kappa q)(\delta+\lambda)B}{\gamma_0\gamma_1 F(z)e^z}\right]^{\frac{1}{\gamma_1-1}}, \qquad B_{\text{fd}}(z,q) = \frac{\gamma_0\gamma_1 F(z)e^z}{(1-\kappa q)(\delta+\lambda)}$$

- No default for $B \leq B_{\text{pd}}(z,q)$, also in closed form. Partial default above it.
- B_{pd} rises with the price: a lower price makes withholding cheaper.
- No-fixed-cost region: $B_{\text{pd}} = 0$, any positive debt is partially defaulted on.
- Regions that discrete-time models find only numerically.

A sovereign never defaults a little

- The intensity is a continuous choice, but the fixed cost makes entry lumpy.
- The first default is a jump of a size set by the costs alone,

$$d_{\text{first}}(z) = \left[\frac{1 - F(z)}{(\gamma_1 - 1)\gamma_0 F(z)}\right]^{1/\gamma_1},$$

  about half of the debt service in the calibrated model ($d_{\text{first}} \approx 0.52$, below).
- Independent of debt, the price, and preferences.

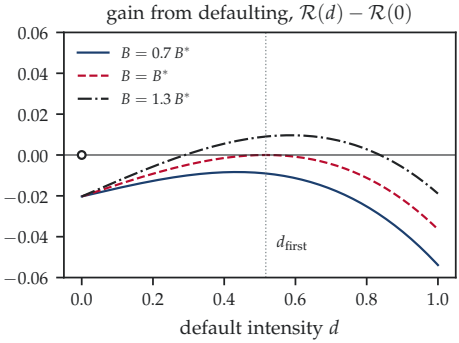
Why the jump

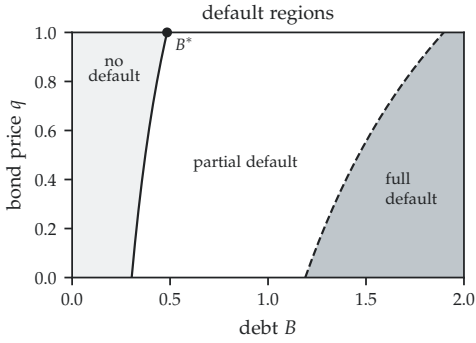
- Without a fixed cost: zero marginal cost at $d = 0$, so a little default at every debt.
- The fixed cost is what makes full repayment possible.
- Once paying the fixed cost, the sovereign withholds until the variable cost bites.
- A steep variable cost (γ_1 large) makes the jump large.

What triggers the jump needs uncertainty

- Not borrowing into default: a shock that moves the threshold below the debt held.
- The chain model adds the one missing element, precaution.

The default regions





- Left: the gain jumps down at $d > 0$, the fixed cost. At B^*, a tie.
- Right: the no-default region shrinks as the price falls. Its edge at q^{rf} is capacity.

The Deterministic Model

The deterministic model

With z fixed at 0, the equilibrium is a pair of ordinary differential equations,

$$V'(B) = \frac{\rho V(B) - u(c^*)}{S^*}, \qquad q'(B) = \frac{\xi(d^*)q(B) - (1-d^*)(\lambda+\delta)}{S^*},$$

with $d^* = d^*(B, q(B))$ static, $c^* = (u')^{-1}(-V'/q)$, and $S^* = (c^* - \mathcal{R}^*)/q$.

- Singular where the drift vanishes: the steady states organize the solution.
- No-default steady state: debt capacity. Partial-default steady state: self-fulfilling.
- Illustrations use the calibrated parameters with the endowment at its mean.

Debt capacity

Theorem (debt capacity in closed form)
In the deterministic model the impatient sovereign borrows at the risk-free price up to

$$B^* = B_{\text{pd}}(0, q^{\text{rf}}),$$

the largest debt it is willing to service in full at the risk-free price, and stops there.

- Closed form in the default costs, the recovery on arrears, and debt maturity.
- Independent of patience, the curvature of utility, and the borrowing limit.

With shocks: the same closed form, at a price below par.
Preferences enter through that price.

Why the sovereign stops at B^*

- Below B^*: no default, price at par, impatience pushes debt up.
- One more unit past B^* starts a default of size d_{first}.
- Lenders would charge for that default at once: the price leaves par.
- The sovereign prefers to stop exactly at the edge of the no-default region.

Contrast with the binary model

- There, the limit is where staying in the market stops covering repayment.
- It depends on patience, risk aversion, and the punishment after default.
- It is found only numerically.

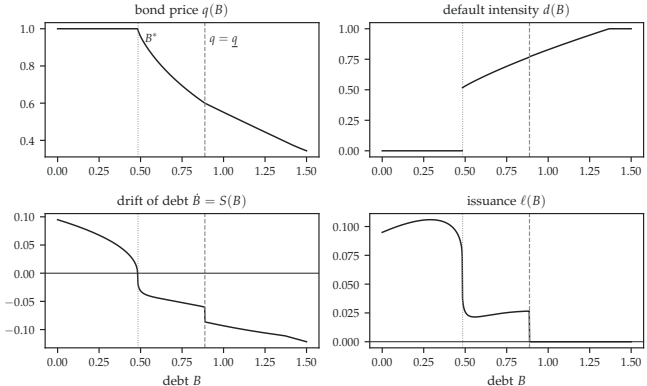
What moves capacity

- **Default costs** γ_0, $1 - F$, and **income**: higher, more capacity.
- **Recovery** κ: more arrears per unit withheld make withholding costlier.
- **Debt service** $\delta + \lambda$: a larger flow to withhold from, lower capacity.

What does not move it

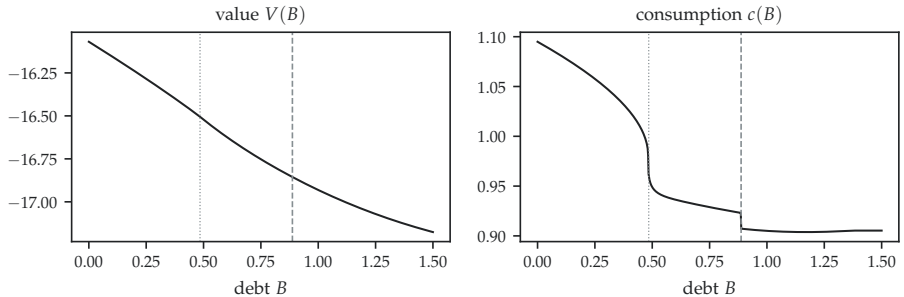
- The discount rate ρ, as long as $\rho > r$. The curvature of u.
- The drift cap and the borrowing limit.
- Comparative statics confirmed on computed solutions (appendix).

The deterministic equilibrium



- Par price and positive drift up to B^*. Partial default and negative drift beyond it.

Value and consumption



- V is C^1 at B^*. Consumption drops steeply but continuously at first default.

Laws of motion around capacity

Lemma (square-root law, from below)

$$S(B^* - x) = (2(\rho - r)\,\psi(\bar{c}))^{1/2}\, x^{1/2}(1 + o(1)), \qquad \psi = u'/|u''|,$$

and B^* is reached in finite time.

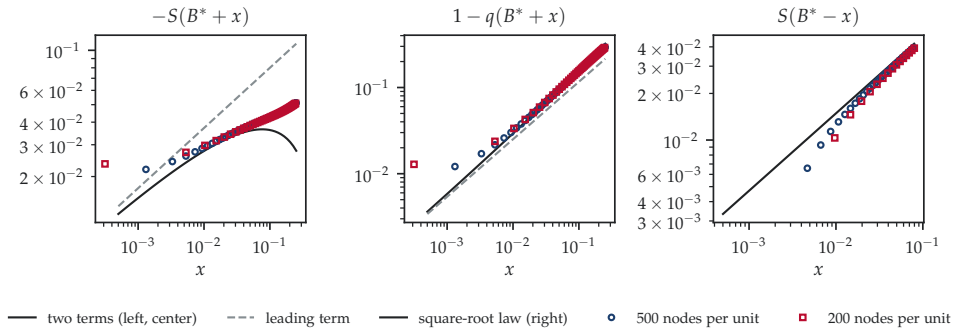
Theorem (cube-root law, from above)

With $\Pi_0 = (1-\kappa)(\delta+\lambda)d_{\text{first}}$ and $\beta = (3a(d_{\text{first}})B^*\Pi_0\,\psi(\bar{c}))^{1/3}$,

$$S(B^* + x) = -\beta\, x^{1/3}(1 + o(1)), \qquad q^{\text{rf}} - q(B^* + x) = \tfrac{3\Pi_0}{2\beta}\, x^{2/3}(1 + o(1)).$$

- The value is C^1 at B^*. Second-order terms in closed form too.

The cube-root law at work



- Repayment slows near capacity. The price leaves par with a vertical tangent.
- Return in finite time: 0.4 years from an excess of 1% of output, 1.9 from 10%.
- Priced at once, worked off without restructuring. The binary model jumps instead.

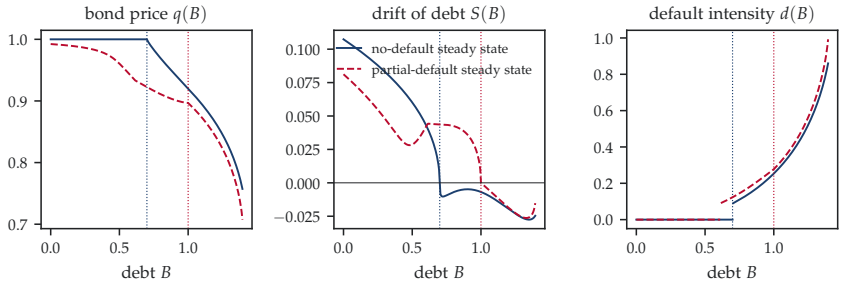
Pessimism alone can produce a slow crisis

- Lenders expect the sovereign to settle at B°, above capacity.
- They expect it to withhold a constant share d° there forever.
- So they pay $q_r(d^\circ) = (1 - d^\circ)(\lambda + \delta)/\xi(d^\circ)$ at B°, and less than par below.

Two consequences validate the expectation

- At the lower price withholding is cheaper, so default starts earlier.
- The default is priced, so the sovereign borrows past its old limit at the risk-free rate.
- It stops at B°, where impatience balances the dilution of its price.

Two equilibria at the same parameters



- Steady states: no default at $B = 0.7$, partial default at $B = 1$ ($d = 28\%$, $q = 0.9$).
- One of a continuum: steady states from 0.93 to 1.02. No run, no loss of access.

When is pessimism self-fulfilling?

Proposition (risk-neutral sovereign)

A self-fulfilling partial default is reached from zero debt only if $\rho - r \geq \Theta^*$: the wedge must cover the output cost of the cheapest default per unit of funds raised. When the default cost is not too convex,

$$\Theta^* = s_{\text{first}},$$

the spread implied by withholding the first default intensity forever. Equilibria exist for a range of wedges above it.

- In the example: $s_{\text{first}} = 47$ bp, stable up to 80 bp.

The discrete-issuance test

- At zero debt, issue the steady-state debt at once and collect its market value.
- From then on: pay the risk-free rate on what was raised, the default is priced.
- And lose output $1 - \phi(d, 0)$ to the default, forever.

Front-loading vs. output cost

- Gain: the impatience wedge times the funds raised, per unit of time.
- Pessimism is self-fulfilling when the gain covers the output cost.
- At the first default, that cost equals the spread times the value of the debt.

Unlike a rollover crisis, the sovereign can stay out of the default region.

Comparison with slow-moving and rollover crises

- **Lorenzoni–Werning (2019):** pessimism raises the cost of rollover, debt drifts.
- **Here:** pessimism raises the default intensity, which justifies the price.
- **Calvo (1988):** expected repudiation raises rates, which justify repudiation.
- **Cole–Kehoe (2000):** a run. Here: no run, no loss of market access.
- **Aguiar et al. (2022):** closest, the government keeps issuing at depressed prices.

The difference: no forced rollover. The sovereign must choose to borrow into default.

The Chain Model

The chain model

The log endowment takes values z_j and switches from j to k at rate η_{jk}.

$$\rho V_j(B) = \max_{c,d} \left\{ u(c) + S\, V_j'(B) + \textstyle\sum_{k \neq j} \eta_{jk} \left[V_k(B) - V_j(B) \right] \right\},$$
$$\xi(d_j)\, q_j(B) = (1 - d_j)(\lambda + \delta) + \tilde{S}_j\, q_j'(B) + \textstyle\sum_{k \neq j} \eta_{jk} \left[q_k(B) - q_j(B) \right].$$

- The static default decision and the thresholds hold state by state, at $q = q_j(B)$.
- One new element: precaution. It can hold a steady state short of capacity.
- The bridge between the deterministic model and the diffusion.

Precaution

Endowment on a finite chain, switching from j to k at rate η_{jk}.

Proposition (steady states at the risk-free price)

At an interior steady state B° of state j in a default-free set,

$$(\rho - r)\, u'(c_j) = \sum_{k \neq j} \eta_{jk} \left(u'(c_k) - u'(c_j) \right).$$

The state with the lowest consumption has no such steady state.

- Impatience gain from borrowing = precautionary loss from a worse state.
- The worst state borrows to its edge. Better states can hold a buffer.

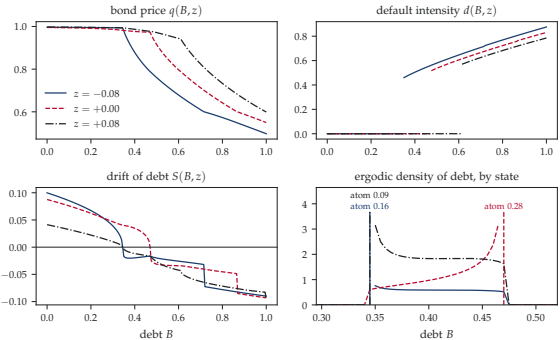
The price kink and when defaults occur

- Past a worse state's threshold, a switch means immediate default: a price kink.
- The kink can hold the better state at or below the worse state's threshold.
- Then a switch does not land the sovereign in default.

Default occurs when a shock moves the threshold below the debt

- Chain: a rare jump. Diffusion: the threshold moves, and the sovereign is caught.
- How often: the tail of shocks against the buffer, not borrowing into default.

A three-state illustration



- $z \in \{-0.08, 0, 0.08\}$, switches at rate 0.1, rare jumps to the low state at rate 0.03.
- The middle state stops at its threshold, the high state at the low state's kink.
- Default on 0.07 of the mass, all in the low state. Debt falls during it.

The Diffusion Model

The diffusion model

Ornstein–Uhlenbeck log endowment, $dz_t = -\mu z_t\,dt + \sigma\,dW_t$, reflected at $\pm\bar{z}$:

$$\rho V = \max_{c,\,d} \left\{ u(c) + S\,V_B - \mu z\,V_z + \tfrac{1}{2}\sigma^2 V_{zz} \right\},$$

$$\xi(d)\,q = (1-d)(\lambda+\delta) + \tilde{S}\,q_B - \mu z\,q_z + \tfrac{1}{2}\sigma^2 q_{zz}.$$

- The continuous-time counterpart of the annual AR(1): $\mu = -\ln \rho_z$.
- Stationary density from the adjoint of the same operator.
- Thresholds move continuously with the endowment.

Where the theory shows: regions and capacity in the policies, stationary debt in the ergodic set, the fixed cost in the response to a shock.

Calibration strategy

- Ornstein–Uhlenbeck endowment, $dz = -\mu z\, dt + \sigma\, dW$. CRRA utility, $\nu = 2$.
- Targets, free parameters, and weights of Arellano, Mateos-Planas, and Ríos-Rull.
- Nine free parameters: default costs, discount rate, recovery, maturity, endowment.
- Eleven targets: default frequency and depth, debt, debt service, spreads, output.
- All nine parameters by direct search: a solution takes minutes.
- Underwriting standard at $\underline{q} = 0.6$.

Parameters

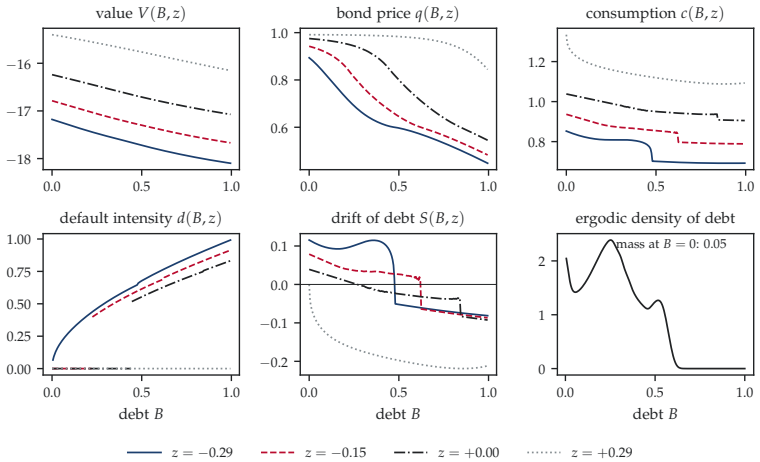
		AMR	This paper
Fixed			
ν	risk aversion	2	2
r	risk-free rate ($\ln 1.04$)	0.039	0.039
q	underwriting standard	–	0.60
Calibrated			
γ_0	variable default cost, level	0.0476	0.0759
γ_1	variable default cost, curvature	2	3.07
γ_2	fixed default cost, slope	0.12	0.082
y^*	fixed default cost, threshold (level)	0.938	0.751
ρ	discount rate (factor $e^{-\rho}$)	0.047	0.062
κ	recovery of defaulted coupons	0.7	0.373
δ	maturity rate (survival $1-\delta$)	0.12	0.153
ρ_z	annual persistence of the endowment	0.875	0.819
σ_ϵ	annual innovation s.d.	0.052	0.066

Fit

Moment	Weight	Data	AMR	Model
Targeted				
Partial default: frequency	1	0.36	0.37	0.38
Partial default: mean $\mid d > 0$	1	0.38	0.39	0.41
Partial default: s.d. $\mid d > 0$	1	0.22	0.19	0.22
Debt to output: mean	1	0.32	0.32	0.27
Debt to output: s.d.	1	0.18	0.25	0.19
Debt service to output: mean	1	0.036	0.035	0.037
Debt service to output: s.d.	1	0.021	0.022	0.020
Debt due to output: mean	3	0.049	0.050	0.052
Spread: s.d.	1.25	0.041	0.037	0.035
Output: persistence	1	0.89	0.88	0.88
Output: s.d.	1	0.10	0.12	0.11
Loss			0.228	0.096
Untargeted				
Defaulted coupons to output: mean $\mid d > 0$		0.052	0.040	0.041
Defaulted coupons to output: s.d. $\mid d > 0$		0.064	0.036	0.028
Spread: mean		0.053	0.016	0.029
Spread: correlation with output		−0.17	−0.38	−0.82
Spread: correlation with debt		0.24	0.56	0.88
Consumption s.d. relative to output s.d.		1.00	0.91	0.87

Matches the frequency and depth of default, debt service, debt due, and output.
Less debt and a lower spread s.d. than the data. Spreads track the state tightly.

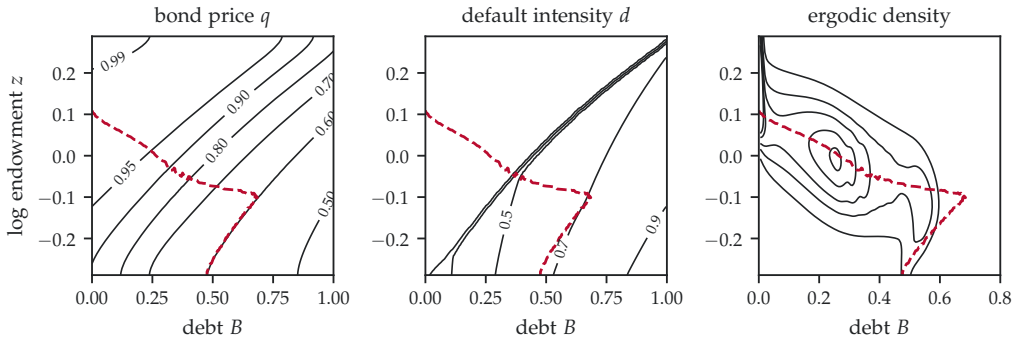
Values, policies, and prices



Reading the policies

- The default policy follows the closed-form rule at every endowment.
- Lowest endowment (no fixed cost): default at any positive debt.
- Average and high endowments: no default until $B_{\text{pd}}(z, q)$, then a jump to d_{first}.
- The threshold rises with the endowment. Full default never reached.
- Consumption falls at the threshold, nearly flat in debt beyond it.
- Spreads low most of the time, high in the tail.

The ergodic set



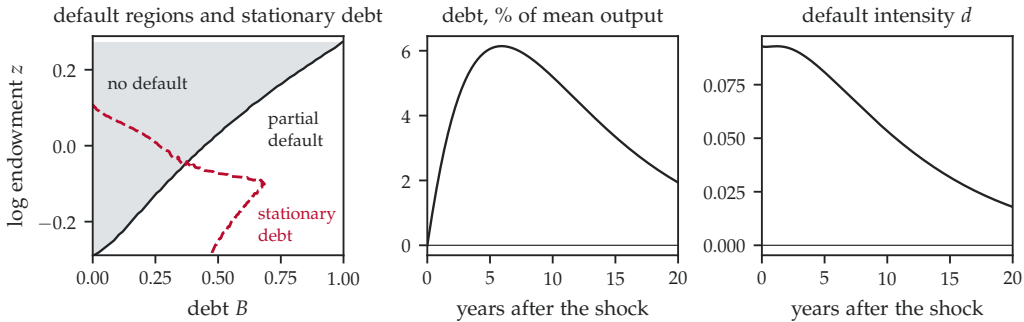
- No-default region: a wedge in (B, z). Mass concentrates along the zero-drift locus.

Stationary debt across states

- **Good times:** stationary debt below the no-default threshold, held by precaution.
- **Bad times:** threshold below the debt. Partial default, fixed cost sunk.
- Stationary debt set by the dilution of the sovereign's own price, inside the default region.
- **Worst states:** stationary debt where the price hits the floor and issuance stops.
- **Very good times:** the sovereign repays all the way to zero debt.

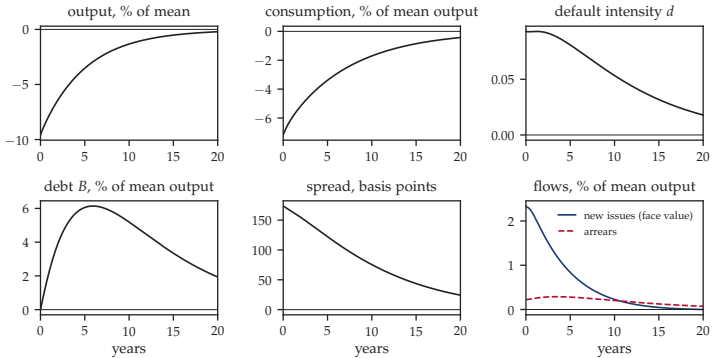
The locus rises from good to bad times, peaks at $B \approx 0.67$, and bends back to 0.48. Default is lumpy at entry and sticky thereafter.

The anatomy of a crisis



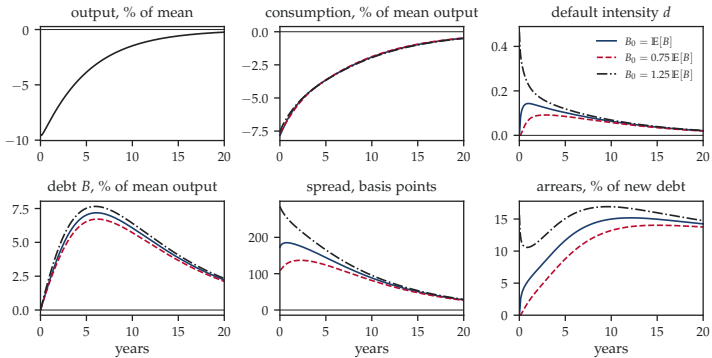
- In bad times the fixed cost is paid regardless and no longer restrains borrowing.
- Stationary debt set by dilution of the sovereign's own price, higher than in good times.

After a bad shock



- Shock smoothed by issuing and withholding.
- To the peak: issues add 8.5, arrears 1.6 (% of output).
- Episode ends when output recovers, not when debt is repaid.

The same shock from three states



- Mean endowment; debt at its ergodic mean, 25% below, 25% above.
- Debt responds the same way from all three.
- Default and the spread depend on the state: from above, the jump at once.
- Arrears about 15% of new debt.

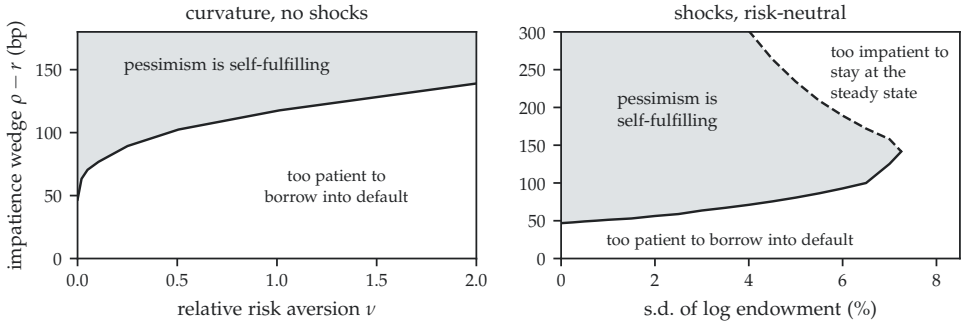
What the episode explains

- Default jumps on impact and stays elevated for a decade.
- Debt rises for about six years, by some six percent of output.
- The spread rises by about 174 bp on impact and declines with recovery.
- New issues exceed defaulted coupons in face value for five and a half years.

The facts of partial default episodes

- Default is protracted. Debt rises during the episode.
- Countries exit owing more than when they entered.
- Market access is the larger cushion in the first years. Arrears are the smaller.

Curvature and uncertainty raise the bar



- Risk aversion of two roughly triples the impatience needed (139 vs. 47 bp).
- Shocks: window narrows from both sides, closes at endowment s.d. $\approx 7\%$.

Why curvature and shocks work against it

Curvature

- A sovereign that values smooth consumption gains less from front-loading.
- It cannot make a discrete issuance. It drifts into default gradually.
- The threshold rises with risk aversion and approaches s_{first} only slowly.

Uncertainty (risk-neutral, steady debt independent of the endowment)

- From below: in bad states, a deeper and costlier default at the steady state.
- From above: the steady state must be stable in every state.
- Waiting at zero debt for a better endowment does not help.

Self-fulfilling default at the calibrated parameters

- Stationary spread at the first default: 1,286 bp. Spread at the floor: 1,278 bp.
- The standard excludes partial-default steady states.
- Impatience wedge: 225 bp, against a risk-neutral threshold of 1,286 bp.
- Curvature and shocks raise the bar further.
- A second equilibrium exists: default on the last 0.3% of borrowing, 18 bp (appendix).

The calibrated sovereign defaults because of shocks, not because of pessimism.

Conclusion

Takeaways

- Arrears raise funds at a price fixed by the recovery. The haircut falls on creditors.
- Debt capacity is a closed form in the cost of defaulting and the bond price.
- Crises begin with a jump, when shocks overwhelm a precautionary buffer.
- The episode is a financing regime: borrowing continues until output recovers.
- Pessimism can cause a slow crisis without a run. Curvature, shocks raise the bar.

Tractable and quantitative at once: fits AMR's targets at least as well as their model.
No smoothing shocks, exact moments and responses, minutes per solution.

Robustness

Robustness of the calibrated point

Grid	Debt/y mean	Debt/y s.d.	Spread s.d.	PD freq.	PD mean	Loss	Mass top	Unresolved
main grid	0.273	0.189	0.035	0.378	0.409	0.096	0.0000	0.015
debt step halved	0.272	0.188	0.035	0.376	0.410	0.096	0.0000	0.007
non-uniform debt grid	0.273	0.188	0.035	0.375	0.410	0.096	0.0000	0.005
barrier at $B = 1.5$	0.273	0.189	0.035	0.378	0.409	0.096	0.0000	0.015
endowment range ± 3 s.d.	0.273	0.186	0.035	0.379	0.406	0.107	0.0000	0.015
endowment range ± 3.5 s.d.	0.272	0.185	0.035	0.378	0.404	0.110	0.0000	0.014

- Finer, non-uniform, and wider debt grids: moments and loss unchanged.
- Wider endowment range: debt, default, and spread moments unchanged. The loss rises with the s.d. of output.

Comparative statics of capacity

Variation	Steady state $B_{\text{pd}}(0, q^{\text{rf}})$	Steady state computed	First default d_{first}	First default computed	$B_{\text{fd}}(0, q^{\text{rf}})$	Standard binds from	Full default from
Calibrated point	0.485	0.475	0.517	0.519	1.90	0.890	–
$\gamma_0 \times 0.8$	0.451	0.440	0.556	0.557	1.52	0.785	1.11
$\gamma_0 \times 1.2$	0.514	0.505	0.487	0.489	2.28	0.990	–
$\kappa = 0.45$	0.553	0.540	0.517	0.518	2.16	1.005	–
$\kappa = 0.65$	0.869	0.850	0.517	0.517	3.40	–	–
$\gamma_2 \times 0.5$	0.305	0.295	0.411	0.412	1.92	0.710	–
$\gamma_2 \times 1.5$	0.635	0.625	0.592	0.594	1.88	1.040	–
$\rho = 0.05$	0.485	0.475	0.517	0.519	1.90	0.940	–
$\rho = 0.07$	0.485	0.475	0.517	0.519	1.90	0.860	–
$\delta = 0.12$	0.584	0.575	0.517	0.519	2.28	1.020	–
$\delta = 0.22$	0.358	0.350	0.517	0.520	1.40	0.705	1.03

- Closed form vs. computed steady state: within a grid step or two in every case.
- Capacity rises with γ_0, κ, γ_2, falls with δ, does not move with ρ.

Other Material

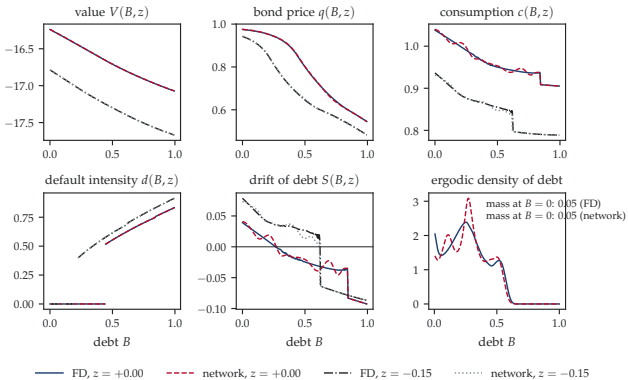
Computation: the scheme

- Implicit upwind finite differences (Achdou et al. 2022), 201×75 grid on (B, z).
- Value and price updated jointly each step. Alternating between them is unstable.
- At each node, three candidates: forward, backward, and zero drift.
- For each: closed-form drift, golden-section search over d, plus $d \in \{0, 1, d_{\text{int}}\}$.
- Keep the candidate with the largest Hamiltonian: monotone upwinding.
- Handles the fixed cost's non-concavity directly. No taste shocks.

Computation: accuracy and speed

- Max residuals: HJB 9×10^{-7}, pricing 10^{-3} (locus), 2×10^{-5} elsewhere.
- Stationary distribution: the transposed generator, solved directly.
- Impulse responses: the forward equation on the shifted distribution. No sampling.
- One parameter evaluation: about a minute on 25 endowment nodes, four on 75.
- Deterministic model: an exact reverse-shooting solution of the discrete system, as a check.

A neural-network solution of the same model



- One network $(B,z) \mapsto (V,q)$, 128×3 tanh; derivatives by autodiff; the same policy kernel at random points.
- Minimizing the stationary residuals fails: wrong drift over much of the state space.
- Trained instead by the same implicit pseudo-time iteration as the grid scheme, on Sobol points.
- No grid, so the cost does not grow with the number of states.

Finite differences vs. the network

	Finite differences	Neural network
Debt to output, mean	0.273	0.272
Debt to output, s.d.	0.189	0.183
Spread, s.d.	0.035	0.034
Partial default frequency	0.378	0.377
Partial default, mean	0.409	0.403
Spread, mean	0.029	0.028
Calibration loss	0.096	0.102
Mass at zero debt	0.053	0.052
Value V, rms / max difference	–	0.003 / 0.008
Bond price q, rms / max difference	–	0.002 / 0.006
First-default threshold, mean / max difference	–	0.001 / 0.005
Drift sign agreement (percent)	–	98.2

- Thresholds within a grid step; mean debt 0.257 vs. 0.257; loss 0.102 vs. 0.096.
- Largest differences at the kinks of the price: the floor locus and the stationary debt.

High debt and the underwriting standard

- Without a limit, debt cannot stay above every full-default threshold forever.
- With a limit and no standard, the sovereign borrows up to it, whatever its level.
- With the standard: debt shrinks in full default, the full-default region is transient.
- Any limit above the full-default thresholds is an exact truncation.
- Where it binds, default follows the closed-form rule at the shadow price $-V_B/u'(c)$.
- Unconstrained full-default issuance price ≈ 0.3, far below the floor of 0.6.

The price and the value

- Price in $[0, q^{\mathrm{rf}}]$: q^{rf} only on default-free paths, 0 only in full default forever.
- The price can jump only where debt stops and is not reached from both sides.
- Full default forever, at a zero price: a knife edge, ruled out by no trade at $q = 0$.
- If the price falls with debt, so does the value, with buybacks and the standard.

Local conditions and the continuum

Lemma (at a partial-default steady state)
The price is $q^\circ = q_r(d^\circ)$, and the slope of the price is pinned by impatience,

$$-\frac{q'(B^{\circ+})}{q^\circ} = \frac{\rho - r}{a(d^\circ)\, B^\circ}.$$

- One candidate for every intensity $d^\circ \geq d_{\text{first}}$: a continuum.
- Stable only if more debt dilutes the price enough, through more default.
- Equilibrium depends on the whole path: borrowing into default from zero debt.
- Lenders' expectations alone select among the candidates.

Self-fulfilling default with shocks: the construction

Risk-neutral sovereign, diffusion endowment, expected steady debt B° in every state.

- At B°: $\xi(d^\circ)q^\circ = (1-d^\circ)(\lambda+\delta) + \mathcal{L}_z q^\circ$, an ODE in z.
- Above B° the sovereign is indifferent, $V_B = -q$, and the price solves

$$a(d)B\,q_B = (1-d)(\lambda+\delta) - (\rho + a(d))\,q + \mathcal{L}_z q.$$

- Below B°: a discrete issuance at q°. It must beat staying debt-free in every state.
- Checks: stability, continuation, the floor. At $\sigma = 0$, exactly the deterministic test.